

NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Department of Mathematics & Computing

Mid Semester Examinations II

MC-A IV Semester

MAPC 204: Computer Oriented Numerical Methods

Date: 08-04-2026

Time: 11.00 AM - 11.50 AM

Max. Marks: 15

Answer the entire question at one place. State all the theorems and results used.

- Q1. The following table gives the distance $d(t)$ (in meters) travelled by a car along a straight road at different times t (in seconds):

time t	0	2	3	5	6	8	10	11	13	14
Distance $d(t)$	0	2	8.7	48.5	87.6	219.2	442	595.1	999.7	1257.2

Estimate the speed and acceleration of the car at $t = 8$ seconds using appropriate numerical differentiation methods. [3]

- Q2. Use Picard's method to approximate the solution of the following IVP [4]

$$y' = e^{t-y}, \quad 0 \leq t \leq 1, \quad y(0) = 1 \quad \text{with} \quad h = 0.5.$$

- Q3. Compute by Gaussian quadrature [4]

$$I = \int_0^1 \frac{\log(x+1)}{\sqrt{x(1-x)}} dx,$$

for $n = 2, 3$.

- Q4. Determine a, b and c such that the relation [4]

$$f'(x_0) = af(x_0) + bf(x_1) + cf(x_2) + R, \quad x_j = x_0 + jh$$

is exact for polynomials of as high degree as possible. Find the error term R .

Max. Marks- 15

Note: Attempt any three questions. Each question carries 5 marks each.

Ques-1 (a) Given a disk queue with the following request sequence: 98, 183, 37, 122, 14, 124, 65, 67. Initial head position = 53. Calculate the total head movement using the following algorithms:

- a) FCFS
- b) SSTF
- c) SCAN (direction toward 0)
- d) C-SCAN

(2.5 marks)

Ques-1 (b) Given five memory partitions of sizes: 100 KB, 500 KB, 200 KB, 300 KB, 600 KB (in order). Processes arrive in the order: 212 KB, 417 KB, 112 KB, 426 KB. Show how the following algorithms allocate memory and which algorithm makes the most efficient use of memory? Justify your answer:

- First-Fit
- Best-Fit
- Worst-Fit

(2.5 marks)

Ques-2 (a) Compare internal fragmentation and external fragmentation in contiguous allocation. Explain how compaction helps reduce fragmentation.

(2.5 marks)

(b) Why is random access inefficient in linked allocation? Explain how indexed allocation overcomes the limitations of contiguous and linked allocation.

(2.5 marks)

Ques-3 On a system using paging and segmentation, references to swapped-in locations accessible through an entry in an associative table take 150 ns. If the main memory page table must be used, the reference takes 400 ns. References that result in page faults require 8 ms if the page to be replaced has been

modified, 3 ms otherwise. If the page fault rate is 2%, the associative table hit rate is 70%, and 50% of replaced pages have been modified, what is the effective access time? Assume the system is running only a single process and the CPU is idle during page swaps. (5 marks)

Ques-4 A page-replacement algorithm should minimize the number of page faults. We can achieve this minimization by distributing heavily used pages evenly over all of memory, rather than having them compete for a small number of page frames. We can associate with each page frame a counter of the number of pages associated with that frame. Then, to replace a page, we can search for the page frame with the smallest counter.

a. Define a page-replacement algorithm using this basic idea. Specifically, address these problems:

- (i) What the initial value of the counters is?
- (ii) When counters are increased?
- (iii) When counters are decreased?
- (iv) How the page to be replaced is selected?
- b. How many page faults occur for your algorithm for the following reference string, with four-page frames? 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 8, 9, 7, 8, 9, 5, 4, 5, 4, 2.
- c. What is the minimum number of page faults for an optimal page replacement strategy for the reference string in part b with four-page frames?

(5 marks)